Spatio-Temporally Inconsistent Hyperspectral Image Superresolution via Category-Guided Diffusion With Scale-Co-Occurrence Assistance

Bo Zhu, Wenqian Dong, *Senior Member, IEEE*, Jiahui Qu, *Senior Member, IEEE*, Nan Li, and Yunsong Li, *Member, IEEE*

***Abstract*—Fusion-based hyperspectral image superresolution (HSI-SR) methods have achieved strong performance. However, these methods typically assume that the high-resolution multispectral image (HR-MSI) and low-resolution hyperspectral image (LR-HSI) are captured over the same scene at the same time, which is often impractical. Leveraging spatio-temporally inconsistent HR-MSI to assist LR-HSI in HSI-SR is significant for breaking this constraint. HSI-SR based on such unpaired images faces two challenges: identifying structurally similar texture details from unpaired images without spatial correspondences while coping with insufficient reference information; reliably aggregating spatial details from the reference image while preserving spectral fidelity. To address these challenges, we propose a scale-co-occurrence assisted category-guided matching and aggregation diffusion network (C^2MA-Diff-Net), built upon a conditional diffusion framework to perform matching and aggregation for unpaired HSI-SR. In particular, during the reverse denoising process, we design a novel matching and aggregation block that integrates a self-scale-co-occurrence assisted category-guided patch matching module (C^2PM) and a spectral–spatial corrected aggregation module (S^2CA) to jointly perform patch matching and feature aggregation, progressively refining the noisy latent. The C^2PM module performs category-guided progressive patch matching (CGPM) with the assistance of self-scale-co-occurrence, efficiently identifying texture-similar patches from HR-MSI while mining cross-scale complementary information from LR-HSI itself. The S^2CA module applies center-aware sliding-window spectral–spatial attention to correct reference features and selectively aggregates global information via adaptive-gated Cross-Mamba, enabling reliable aggregation while preserving spectral fidelity. Experimental results show that C^2MA-Diff-Net delivers superior performance on unpaired HSI-SR, confirming the effectiveness of our method. Code is available at https://github.com/Jiahuiqu/C2MA-Diff**

***Index Terms*—Aggregate, diffusion, hyperspectral image superresolution (HSI-SR), match, scale-co-occurrence, spatio-temporally inconsistent.**

I. Introduction

HYPERSPECTRAL images (HSIs) contain continuous and narrow spectral bands, enabling precise characterization of fine-grained spectral signatures of ground objects. However, due to physical limitations of imaging sensors and acquisition conditions, HSIs usually suffer from low spatial resolution, which restricts their performance in downstream tasks such as classification [1], [2], [3], semantic segmentation [4], change detection [5], and anomaly detection [6]. Hyperspectral image superresolution (HSI-SR) aims to reconstruct high-resolution hyperspectral images (HR-HSI) with both high spatial resolution and spectral fidelity from low-resolution observations, and is an important technique to overcome this limitation.

According to whether external auxiliary information is introduced, existing HSI-SR methods can be broadly categorized into two groups: single hyperspectral image superresolution (single HSI-SR) and fusion-based HSI-SR (fusion-based HSI-SR) [7], [8]. Single HSI-SR methods enhance spatial resolution by exploiting the internal spatial correlations of low-resolution hyperspectral images (LR-HSIs) without relying on additional high-resolution images. However, due to the severely illposed nature of the problem, recovering sufficiently high-frequency details from a single observation is extremely challenging, and the reconstruction quality is often limited. On the contrary, fusion-based HSI-SR methods jointly utilize the spectral information of LR-HSI and the spatial details of high-resolution multispectral images (HR-MSIs), enabling cross-modal complementarity and generating HR-HSI with more realistic spatial structures and spectral characteristics [9], [10], [11]. Consequently, fusion-based HR-HSI methods have become the mainstream research direction.

However, as illustrated in Fig. 1, most existing fusion-based HSI-SR methods rely on an ideal assumption that cross-modal image pairs are acquired over the same scene at the same time, ensuring strict spatial–temporal consistency [12]. In practice, due to variations in imaging conditions,

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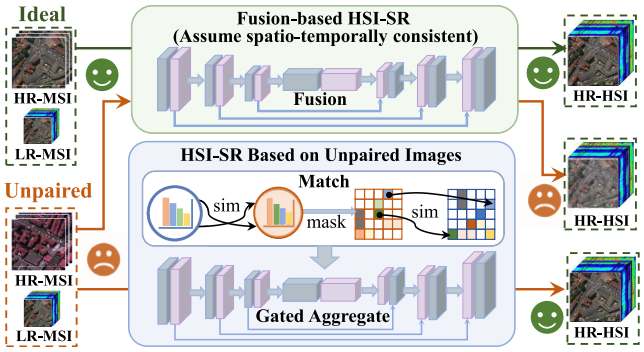


Fig. 1. Existing fusion-based HSI-SR methods predominantly rely on strictly aligned image pairs, whereas our proposed method performs superresolution using unpaired image pairs.

sensor discrepancies, and temporal changes, maintaining such consistency is extremely difficult. Cross-modal image pairs with spatial mismatches caused by differences in acquisition time or scene conditions are referred to as unpaired images. When provided with such unpaired inputs, the performance of existing fusion-based HSI-SR methods degrades significantly. If HSI-SR could be achieved using spatio-temporally inconsistent images that share locally similar textures, it would remove the strict requirement for spatial–temporal consistency and greatly broaden the applicability of HSI-SR methods. This task is referred to as unpaired HSI-SR.

The key to achieving unpaired HSI-SR lies in identifying regions with similar texture structures between unpaired images and transferring useful texture details from these similar regions to the target image. In the field of natural image processing, reference-based superresolution (Ref-SR) methods for RGB images have achieved outstanding performance [13], [14]. These methods utilize patch-based similarity matching to find regions with similar texture structures in high-resolution reference images for subsequent information aggregation and texture transfer, thereby assisting in the reconstruction of high-resolution images from low-resolution inputs. This provides a source of inspiration for unpaired HSI-SR, suggesting that patch matching can be employed to locate similar regions for subsequent texture transfer. As shown in Fig. 1, we adopt a joint matching–aggregation framework for HSI-SR with unpaired images. In particular, a matching module searches for highly similar patches between the target LR-HSI and the reference HR-MSI, while an aggregation module reconstructs HR-HSI by transferring the matched textures.

However, remote sensing scenes are considerably more complex than natural images, with densely distributed categories and intricate boundaries, making cross-image matching prone to mismatches and erroneous texture transfer. Moreover, under unpaired settings, reference images may lack certain categories or structures, making reference information inherently incomplete. Therefore, the first core challenge is to achieve reliable cross-image matching despite semantic ambiguity and incomplete References. The second core challenge is to reconstruct HR-HSI by effectively utilizing reference details while preserving spectral fidelity and global structural coherence.

Diffusion models have recently demonstrated strong capability in modeling complex image distributions and generating high-quality details via iterative denoising [15], making them well-suited for HSI-SR under spatio-temporally inconsistent reference conditions. Their progressive denoising process can tolerate and gradually correct uncertainties and mismatched textures across iterations, leading to more reliable high-frequency detail reconstruction than single-step inference.

To address the above challenges, we propose a scale-co-occurrence assisted category-guided matching and aggregation diffusion network (C^2MA-Diff-Net), which is built upon a conditional diffusion framework to iteratively perform feature aggregation and progressively reconstruct HR-HSI from noise. In particular, during the reverse denoising process, we design a matching and aggregation block that integrates a self-scale-co-occurrence assisted category-guided patch matching module (C^2PM) and a spectral–spatial corrected aggregation module (S^2CA) to jointly perform patch matching and feature aggregation, enabling progressive refinement of the latent representation and reconstruction of HR-HSI with rich spatial details. The C^2PM module performs category-guided progressive patch matching (CGPM) with self-scale co-occurrence assistance. It restricts the matching scope using category information while leveraging scale-co-occurrence to compensate for missing categories in reference images, thereby achieving accurate and efficient matching while supplementing additional details from the image itself. The S^2CA module calibrates reference features and selectively integrates textures via an adaptive-gated Cross-Mamba mechanism.

The main contributions of this work are summarized as follows.

1) We propose C^2MA-Diff-Net, a conditional diffusion-based framework that iteratively performs matching and aggregation for unpaired HSI-SR, effectively removing the strict reliance on spatial–temporal consistency between image pairs and significantly expanding the applicability of HSI-SR methods.
2) To identify similar texture structures in unpaired images, we design a C^2PM module for precise and efficient patch matching. It performs CGPM with self-scale-co-occurrence assistance, enabling accurate matching while exploiting cross-scale self-similarity as supplementary information.
3) To achieve reliable feature aggregation while preserving spectral fidelity, we develop an S^2CA module that calibrates reference features using center-aware sliding-window spectral–spatial attention and selectively integrates textures via an adaptive-gated Cross-Mamba mechanism.

The rest of this article is organized as follows. Section II reviews related work. Section III presents the proposed C^2MA-Diff-Net. Section IV reports comparison and ablation experiments on three constructed datasets. Finally, Section V concludes this article.

II. Related Work

A. Single HSI-SR

Single HSI-SR enhances spatial resolution by exploiting intrinsic spatial–spectral correlations within LR-HSI without

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external [16]. Early methods mainly relied on interpolation and sparse representation, which had limited ability to recover high-frequency details. With the development of deep learning, CNN-based models became dominant. For example, Yang et al. [17] proposed a hybrid 3-D attentive CNN to capture local and nonlocal spectral–spatial dependencies. GAN-based methods [18] improve perceptual quality through adversarial learning, while Transformer-based models such as DSSTSR [19] further model long-range spatial and spectral dependencies.

However, since single HSI-SR relies solely on LR-HSI information, it often struggles to recover sufficient high-frequency details due to the severely illposed nature of the SR problem.

B. Fusion-Based HSI-SR

Fusion-based HSI-SR addresses the limitations of single HSI-SR by exploiting complementary spatial information from HR-MSI. Traditional approaches include component substitution (CS), multiresolution analysis (MRA), Bayesian methods, and matrix factorization, which enhance spatial resolution by injecting HR-MSI spatial details into LR-HSI while preserving spectral consistency.

Recent deep learning methods further improve fusion-based HSI-SR. Dong et al. [20] proposed RHDN, a two-branch CNN with cascade residual hyperdense blocks to model complex feature relationships. Attention-based SSPSR [21] improves feature extraction via spatial–spectral attention, while Transformer-based Fusformer [22] captures long-range dependencies to alleviate spectral distortion. Adversarial approaches such as HPGAN [23] further enhance visual fidelity by restoring sharper edges. Diffusion models progressively reconstruct target outputs from noise via iterative denoising and have demonstrated strong capability for high-quality generation across various tasks. Li et al. [15] proposed a dual conditional diffusion model-based fusion network (DCDMF) that produces high-quality images through iterative denoising rather than one-step fusion, effectively reducing blurring. Dong et al. [24] proposed an interpretable scale-propelled diffusion (ISPDiff) model for HSI-SR that progressively reconstructs images by learning multiscale distributions. HR-HSI generated by such methods is widely applied in downstream tasks such as change detection [25], [26], and target detection [27].

Despite their superior performance, most fusion-based methods assume spatial alignment between LR-HSI and HR-MSI, which is rarely available in practice. To address this issue, some studies incorporate misregistration during reconstruction. Zhou et al. [28] formulated registration and fusion as a two-stage framework, while Qu et al. proposed PSRF-DiffNet [29], a diffusion-based model that progressively refines both processes. However, these methods still assume that the inputs originate from the same scene and acquisition time, where pixelwise correspondences can be established through registration. This assumption differs fundamentally from the unpaired setting considered in this article.

C. Reference-Based Superresolution

In the domain of natural image processing, Ref-SR has emerged as a prominent research direction, which aims to reconstruct high-resolution images by transferring fine-grained texture from external reference images to enhance low-resolution inputs [30]. CrossNet introduces the use of pretrained VGG features to extract image semantics and match similar patches between LR and reference images, enabling more precise texture transfer [31]. TTSR adopts cross-attention to establish semantic correspondences across images [13]. MASA-SR incorporates deformable convolutions to spatially align misregistered reference features and injects textures via adaptive modulation [14]. High-precision semantic correspondence was achieved through multiscale feature enhancement and hierarchical matching optimization [32]. Transformer-based models were also introduced: Cross-Swin [33] uses Swin Transformer and cross-attention for joint matching and fusion, achieving robust alignment in complex scenes.

Ref-SR has achieved impressive results in natural image processing, and in recent years, a few studies have explored reference-based remote sensing superresolution. For example, Dong et al. proposed Ref-Diff [34], which incorporates land surface change priors into a conditional diffusion model to enhance reference utilization in unchanged regions while preserving semantic reconstruction in changed areas. Tu et al. introduced RGTGAN [35], a reference-based gradient-assisted texture-enhancement GAN that transfers texture details from reference images with different viewpoints to superresolve low-resolution inputs. However, these approaches all require the input images to capture the same scene, either at different times or from different viewpoints, which differs from the cross-temporal, cross-scene, and cross-region unpaired cross-modal setting considered in this work.

III. Method

A. Problem Formulation

The LR-HSI $\mathbf{H} \in \mathbb{R}^{h\times w\times C}$ and the HR-MSI $\mathbf{M}_{ref} \in \mathbb{R}^{H\times W\times c}$ are regarded as degraded observations of two latent HR-HSIs, $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{X}_{ref} \in \mathbb{R}^{H\times W\times C}$. The goal is to reconstruct the target HR-HSI $\mathbf{X}$ from its degraded observation $\mathbf{H}$, with the assistance of the unpaired reference $\mathbf{M}_{ref}$. Although $\mathbf{X}$ and $\mathbf{X}_{ref}$ lack spatial correspondence, they share similar land-cover semantics and exhibit similar structural textures in the same category. This relationship can be expressed as follows:

$$\mathbf{M}_{ref} = \mathbf{X}_{ref}\,\mathbf{R} \tag{1}$$

$$\mathbf{H} = \mathbf{B}\,\mathbf{X} \tag{2}$$

$$\mathbf{X} \overset{\text{sim}}{\longleftrightarrow} \mathbf{X}_{ref} \tag{3}$$

where $\mathbf{R} \in \mathbb{R}^{C\times c}$ is the spectral response matrix simulating spectral degradation from hyperspectral to multispectral domains, $\mathbf{B} \in \mathbb{R}^{HW\times hw}$ denotes the spatial degradation operator, which includes blurring and downsampling, and the notation $\mathbf{X} \overset{\text{sim}}{\longleftrightarrow} \mathbf{X}_{ref}$ indicates semantic-level similarity between $\mathbf{X}$ and $\mathbf{X}_{ref}$ in land-cover categories and textures.

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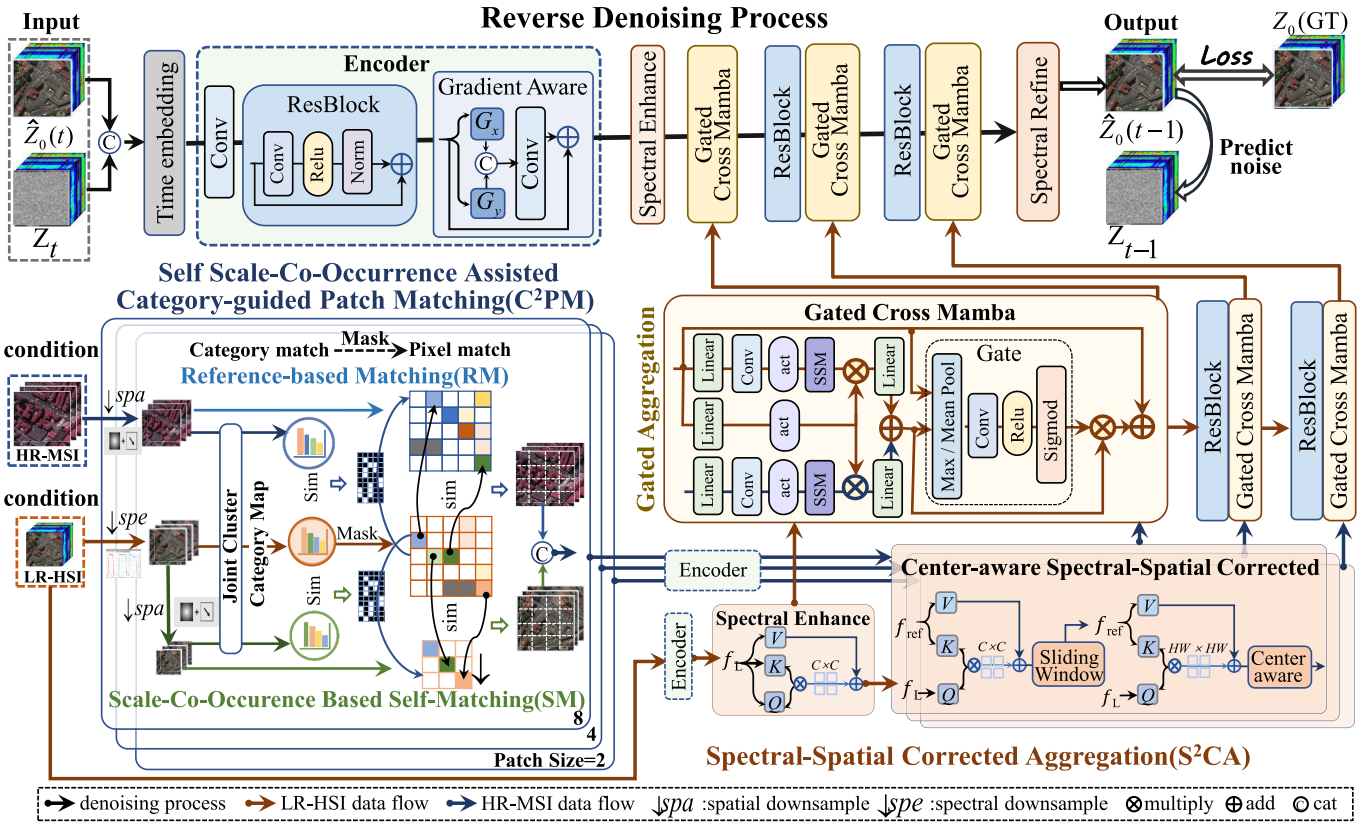


Fig. 2. Overall architecture of the proposed C²MA-Diff-Net for unpaired HSI-SR, built upon a conditional diffusion framework. During the reverse denoising process, C²PM first matches patches between the conditional LR-HSI and HR-MSI to identify structurally similar textures and obtain matched reference images. Subsequently, S²CA progressively aggregates features from the LR-HSI, matched reference images, and noisy HR-HSI to reconstruct the clean HR-HSI.

Motivated by this semantic consistency, we propose to extract and transfer category-consistent textures from $\mathbf{M}_{ref}$ to $\mathbf{H}$ for HR-HSI reconstruction. The reconstruction process is formalized as follows:

$$\mathbf{X} = \mathcal{T}\left(\mathcal{M}^{\mathbf{c}}_{atch}\left(\mathbf{M}_{ref}, \mathbf{H}\right), \mathbf{H}\right) \tag{4}$$

where $\mathcal{M}^{\mathbf{c}}_{atch}(\cdot)$ is a category-guided cross-image similarity matching function, which utilizes a category map $\mathbf{C} \in \mathbb{R}^{h \times w}$ as prior knowledge to constrain patch matching, and $\mathcal{T}(\cdot)$ denotes a feature aggregate function that transfers the matched textures from $\mathbf{M}_{ref}$ into $\mathbf{H}$.

B. Overall Design

As illustrated in Fig. 2, based on the joint optimization of patch matching and information aggregation, we propose a C²MA-Diff-Net built upon a conditional diffusion framework to iteratively perform matching and aggregation for progressive denoising, ultimately achieving unpaired HSI-SR. During the forward process, Gaussian noise is gradually added to generate a pure noise input. In the reverse denoising process, guided by the unpaired LR-HSI and HR-MSI along with their corresponding category maps, the network jointly optimizes patch matching and feature aggregation to generate a predicted HR-HSI for noise estimation. Through iterative denoising, the HR-HSI is progressively reconstructed. In particular, at each denoising step, we design a C²PM and an S²CA to, respectively, achieve category-guided patch matching assisted by scale-co-occurrence and adaptive feature aggregation that takes into account both local and global information.

1) Forward Diffusion Process: The forward diffusion process incrementally adds Gaussian noise to the initial HR-HSI $\mathbf{Z}_0$ over T time steps, gradually transforming it into Gaussian noise. This process is defined by a series of conditional probability distributions, and its overall joint distribution can be expressed as follows:

$$q\left(\mathbf{Z}_{1:T}|\mathbf{Z}_0\right) = \prod_{t=1}^{T} q\left(\mathbf{Z}_t|\mathbf{Z}_{t-1}\right). \tag{5}$$

At each iteration, the conditional distribution is given by the following equation:

$$q\left(\mathbf{Z}_t|\mathbf{Z}_{t-1}\right) = \mathcal{N}\left(\mathbf{Z}_t;\ \sqrt{\alpha_t}\mathbf{Z}_{t-1}, (1-\alpha_t)\mathbf{I}\right) \tag{6}$$

where $\alpha_{1:T} \in (0, 1)$ is a predefined sequence of hyperparameters. In each iteration, the mean term $(\alpha_t)^{1/2}\mathbf{Z}_{t-1}$ controls the retention of information from the previous step $\mathbf{Z}_{t-1}$, while the variance term $(1 - \alpha_t)\mathbf{I}$ introduces noise. Together, they progressively transform the data from clear to blurred, achieving a gradual degradation.

To facilitate subsequent computation and model training, we apply the reparameterization trick to express $\mathbf{Z}_t$ as a function of the initial data $\mathbf{Z}_0$

$$\mathbf{Z}_t = \sqrt{\tilde{\alpha}_t}\mathbf{Z}_0 + \sqrt{1-\tilde{\alpha}_t}\boldsymbol{\epsilon} \tag{7}$$

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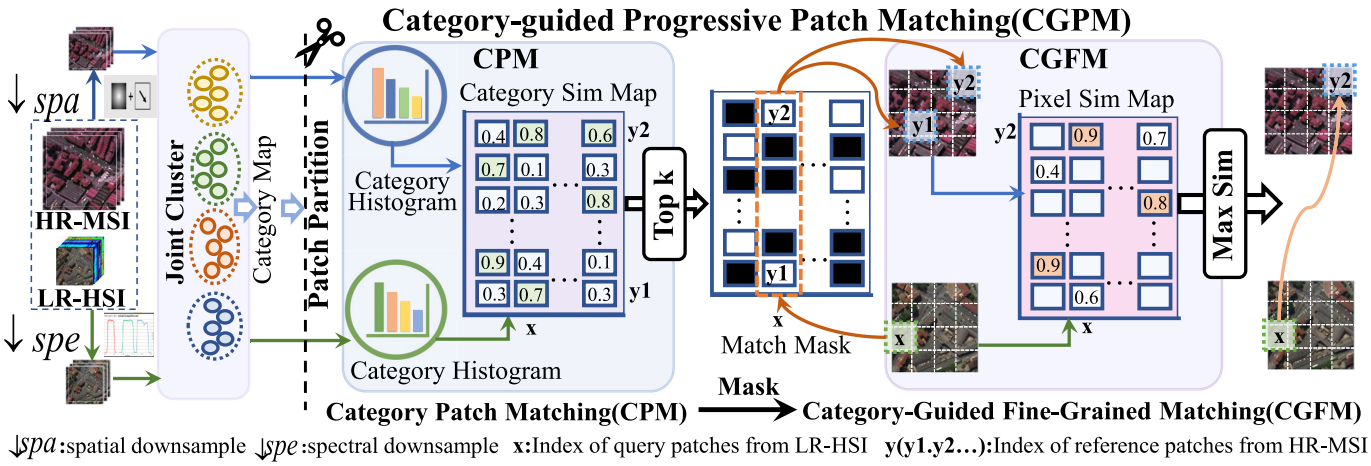


Fig. 3. Overall architecture of the proposed CGPM module. Joint clustering generates category maps, CPM selects category-similar candidate patches, and CGFM further identifies the optimal pixel-level matching patches. The values shown in the maps represent similarity scores.

where $\tilde{\alpha}_t = \prod_{i=1}^{t} \alpha_i$ is the cumulative hyperparameter and $\epsilon \sim \mathcal{N}(0, I)$ is standard Gaussian noise. As the iteration number t increases, $\tilde{\alpha}_t$ gradually approaches 0, meaning the influence of the original data $\mathbf{Z}_0$ in $\mathbf{Z}_t$ diminishes, and ultimately, $\mathbf{Z}_t$ becomes pure Gaussian noise.

2) Backward Category-Guided Matching and Feature Aggregation Process: As shown in Fig. 2, at each step of the backward denoising process, the proposed denoising network p jointly performs matching and feature aggregation of LR-HSI denoted as $\mathbf{H}$ and HR-MSI denoted as $\mathbf{M}_{ref}$ via the C^{2}PM and S^{2}CA modules, to generate an intermediate clean HR-HSI prediction $\hat{\mathbf{Z}}_{0,t-1}$ based on the current noise latent variable $\mathbf{Z}_t$ and the previous prediction $\mathbf{Z}_{0,t}$

$$\begin{aligned}\hat{\mathbf{Z}}_{0,t-1} &= p\left(\mathbf{H}, \mathbf{M}_{ref}, \mathbf{Z}_t, \mathbf{Z}_{0,t}\right) \\ &= \mathrm{S}^2\mathrm{CA}\left(\mathrm{C}^2\mathrm{PM}\left(\mathbf{H}, \mathbf{M}_{ref}\right), \mathbf{H}, \mathbf{Z}_t, \mathbf{Z}_{0,t}\right) \end{aligned} \tag{8}$$

where C^{2}PM performs CGPM between $\mathbf{H}$ and $\mathbf{M}_{ref}$ to obtain the matched reference image $\mathbf{M}_{\mathrm{m}}$ used for reconstruction, and S^{2}CA dynamically aggregates $\mathbf{M}_{\mathrm{m}}$ with features from $\mathbf{H}$, $\mathbf{Z}_t$, and the previous intermediate prediction $\hat{\mathbf{Z}}_{0,t}$ to produce $\hat{\mathbf{Z}}_{0,t-1}$.

Based on the predicted clean image $\hat{\mathbf{Z}}_{0,t-1}$, the corresponding noise $\boldsymbol{\epsilon}$ can be implicitly estimated according to the reparameterization of the forward process. This estimated noise is then used to compute the posterior mean of $\mathbf{Z}_{t-1}$, enabling the reverse diffusion step

$$\mathbf{Z}_{t-1} = \frac{\sqrt{\alpha_t}\left(1-\bar{\alpha}_{t-1}\right)}{1-\bar{\alpha}_t}\mathbf{Z}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\left(1-\alpha_t\right)}{1-\bar{\alpha}_t}\hat{\mathbf{Z}}_{0,t-1} + \sigma_t\boldsymbol{\epsilon}. \tag{9}$$

This noisy image $\mathbf{Z}_{t-1}$, along with the previous prediction $\hat{\mathbf{Z}}_{0,t-1}$, is used as input for the next step of the backward iteration, ultimately achieving a progressive restoration of HR-HSI from the noisy image.

To supervise training, an L_1 loss is employed

$$\mathcal{L}_{L1} = \left\|\hat{\mathbf{Z}}_{0,t-1} - \mathbf{Z}_0\right\|_1 \tag{10}$$

where $\mathbf{Z}_0$ is the ground truth (GT) HR-HSI.

C. Self-Scale-Co-Occurrence Assisted Category-Guided Patch Matching

For unpaired HSI-SR, achieving accurate matching is a prerequisite for subsequent feature aggregation. To achieve this, we propose a C^{2}PM for accurate and reliable patch matching. As shown in Fig. 2, the module adopts a dual-branch architecture that integrates reference-based matching (RM) and scale-co-occurrence based self-matching (SM). In particular, the RM branch performs CGPM between the target LR-HSI and the reference HR-MSI. Considering the potential absence of certain object categories in the reference HR-MSI, the SM branch matches the target LR-HSI with its downsampled version based on the principle of scale-co-occurrence, thereby extracting complementary information from similar and more detailed regions in the LR-HSI. The downsampling factor used for self-matching is aligned with the spatial resolution ratio between the HR-MSI and LR-HSI. Both branches employ our CGPM module to progressively match patches. As depicted in Fig. 3, the CGPM architecture comprises a category patch matching (CPM) module and category-guided fine-grained matching (CGFM) module. The CGPM performs CGPM fully utilizing category information to improve matching accuracy and reduce computational cost. Ultimately, the C^{2}PM module yields complete and accurate matching results to facilitate effective information aggregation among spatially similar image patches.

We illustrate the two-step matching mechanism of the CGPM module using the RM branch as an example.

1) Category Patch Matching: CPM calculates interpatch category similarity to select semantically consistent patches, reducing cross-category mismatches while minimizing computational burden for subsequent processing. In particular, to address modality discrepancies, we apply spectral degradation on the target $\mathbf{H}$ and spatial degradation on the reference $\mathbf{M}_{\mathbf{ref}}$ to generate the low-resolution multispectral images $\mathbf{M}$ and $\mathbf{M}'_{\mathbf{ref}}$. Joint clustering is performed on $\mathbf{M}$ and $\mathbf{M}'_{\mathbf{ref}}$ to generate category maps $\mathbf{L}$ and $\mathbf{L}_{\mathbf{ref}}$ for subsequent category-guided matching. K-means clustering is conducted in the joint feature space, where the number of clusters is empirically

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determined according to the scene complexity and land-cover characteristics of each dataset. Although the clustering results may not fully correspond to true semantic categories, the generated category maps are only used to constrain the matching search space, making the method less sensitive to imperfect clustering results. Subsequently, $\mathbf{M}$ and $\mathbf{L}$ are partitioned into nonoverlapping patches $\mathbf{P}$ and label patches $\mathbf{T}$, while $\mathbf{M}'_{\mathbf{ref}}$ and $\mathbf{L}_{\mathbf{ref}}$ are divided into overlapping patches $\mathbf{P}_{\mathbf{ref}}$ and label patches $\mathbf{T}_{\mathbf{ref}}$. Moreover, we adopt a multiscale patch partition strategy to enhance the adaptability of matching.

Based on the category label maps, the category histogram of each patch is computed, and the category similarity $\mathbf{S}$ between patches is measured by the normalized histogram difference

$$\mathbf{S}\left(\mathbf{P}_i, \mathbf{P}_j^{\text{ref}}\right) = 1 - \frac{1}{2}\sum_c \left| \frac{\mathbf{N}_i(c)}{\|\mathbf{N}_i\|_1} - \frac{\mathbf{N}_j^{\text{ref}}(c)}{\left\|\mathbf{N}_j^{\text{ref}}\right\|_1} \right| \tag{11}$$

where c denotes the category index, $\mathbf{N}_i(c)$ and $\mathbf{N}_j^{\text{ref}}(c)$ represent the number of pixels belonging to category c in each patch $\mathbf{P}_i$ and $\mathbf{P}_j^{\text{ref}}$ and $\|\cdot\|_1$ denotes L_1-norm.

For each target patch $\mathbf{P}_i$, we select its top-k most similar reference patches $\mathbf{P}_j^{\text{ref}}$ based on $\mathbf{S}(\mathbf{P}_i, \mathbf{P}_j^{\text{ref}})$, and encode this correspondence in a binary mask $\mathbf{M}_a$

$$\mathbf{M}_a(i,j) = \begin{cases} 1, & \text{if } \mathbf{P}_j^{\text{ref}} \in \text{Top}_k\left(S_{\text{cat}}\left(\mathbf{P}_i, \mathbf{P}_j^{\text{ref}}\right)\right) \\ 0, & \text{otherwise} \end{cases} \tag{12}$$

where $\text{Top}_k(\cdot)$ selects the k most similar reference patches.

2) Category-Guided Fine-Grained Patch Matching: CGFM performs fine-grained matching constrained by the coarse category matching mask, achieving precise matching while maintaining computational efficiency.

First, CGFM filters candidate patches using a binary mask $\mathbf{M}_{\text{a}}$ to reduce the search space

$$\mathcal{P}_i^{\text{ca}} = \left\{ \mathbf{P}_j^{\text{ref}} \,\middle|\, \mathbf{M}_{\text{a}}(i,j) = 1 \right\} \tag{13}$$

where $\mathcal{P}_i^{\text{ca}}$ contains reference patches selected for fine-grained matching with target patch $\mathbf{P}_i$.

Next, to enhance robustness against orientation mismatches, we apply four rotation transformations $R(\theta)$ for each $\mathbf{P}_j^{\text{ref}} \in \mathcal{P}_i^{\text{ca}}$ and compute the rotation-enhanced similarity$\mathbf{S}_{\text{rot}}$ with the target $\mathbf{P}_i$

$$\mathcal{S}_{\text{rot}}\left(\mathbf{P}_i, \mathbf{P}_j^{\text{ref}}\right) = \max_\theta \left(\mathcal{S}_{\text{p}}\left(\mathbf{P}_i, R_\theta\left(\mathbf{P}_j^{\text{ref}}\right)\right)\right) \tag{14}$$

where $\theta \in \{0°, 90°, 180°, 270°\}$; R_θ denotes rotating the image patch by θ degrees and S_p represents the similarity between image patches, combining cosine similarity $\mathbf{S}_{\text{cos}}$ and distance-based similarity $\mathbf{S}_{\text{dist}}$ through weighted averaging, which can be defined as follows:

$$\begin{aligned} \mathcal{S}_p\left(\mathbf{P}, \mathbf{P}^{\text{ref}}\right) = \frac{1}{HW}\sum_{u=1}^{H}\sum_{v=1}^{W}\Big[\lambda \cdot \mathcal{S}_{\text{cos}}\left(p_{(u,v)}, p_{(u,v)}^{\text{ref}}\right) \\ + (1-\lambda)\cdot \mathcal{S}_{\text{dist}}\left(p_{(u,v)}, p_{(u,v)}^{\text{ref}}\right)\Big] \end{aligned} \tag{15}$$

where $\lambda \in [0,1]$ controls the balance between similarity measures, (u,v) denotes pixel coordinates, and $H \times W$ represents the patch dimensions. The cosine similarity $\mathcal{S}_{\text{cos}}$ and normalized distance similarity $\mathcal{S}_{dist}$ between pixel-level features p are defined as follows:

$$\mathcal{S}_{\text{cos}}\left(p_{u,v}, p_{u,v}^{\text{ref}}\right) = \frac{p_{u,v}\cdot p_{u,v}^{\text{ref}}}{\left\|p_{u,v}\right\|_2 \left\|p_{u,v}^{\text{ref}}\right\|_2} \tag{16}$$

$$\mathcal{S}_{\text{dist}}\left(p_{u,v}, p_{u,v}^{\text{ref}}\right) = 1 - \frac{\left\|p_{u,v} - p_{u,v}^{\text{ref}}\right\|_2}{\max_{u,v}\left\|p_{u,v} - p_{u,v}^{\text{ref}}\right\|_2 + \epsilon} \tag{17}$$

where $\epsilon > 0$ ensures numerical stability.

Finally, the optimal reference patch is selected through

$$\mathbf{P}_{\text{opt}}^{\text{ref}} = *arg\,max_{\mathbf{P}_j^{\text{ref}} \in \mathcal{P}_i^{\text{ca}}} \mathcal{S}_{\text{rot}}\left(\mathbf{P}_i, \mathbf{P}_j^{\text{ref}}\right). \tag{18}$$

The matched patches $\{\mathbf{P}_{\text{m}}^{\text{ref}}\}$ are then fused to reconstruct the HR-HSI spatially matched with the target HR-MSI. Both the RM and SM branches generate their respective matched reference images $\mathbf{M}_{\text{m}}^{\text{ref}}$ and $\mathbf{M}_{\text{m}}^{\text{self}}$ through the CGPM. These outputs are then concatenated to form $\mathbf{M}_{\text{m}}$, which serves as the matched reference image and is fed into the subsequent aggregation network.

Since CPM adopts a top-k selection strategy to choose k representative patches from candidate reference patches for subsequent pixelwise matching, the computational complexity of CGFM is reduced from $O(N_l \times N_h \times H \times W \times C)$ to $O(N_l \times k \times H \times W \times C)$, thereby significantly reducing the computational cost of fine-grained matching. N_l and N_h, respectively, denote the number of image patches obtained by dividing the LR-HSI and HR-MSI.

D. Spectral–Spatial Corrected Gated-Aggregation

To achieve reliable feature aggregation while preserving spectral fidelity, we design the S^{2}CA module to effectively transfer texture details from the matched reference $\mathbf{M}_{\text{m}}$ to the target $\mathbf{H}$. As shown in Fig. 2, the module first performs local spectral–spatial correction and then conducts iterative global selective feature transfer. In particular, the proposed center-aware spectral–spatial correction (CS^{2}C) module mitigates modality discrepancies and patch-level matching errors by aligning $\mathbf{M}_{\text{m}}$ with $\mathbf{H}$ through spectral–spatial correction. It first enhances the spectral representation of $\mathbf{H}$ via spectral self-attention, then calibrates $\mathbf{M}_{\text{m}}$ using spectral cross-attention, followed by center-aware spatial correction to refine local spatial correspondence within sliding windows. Finally, the proposed gated aggregation (GA) module selectively aggregates the corrected reference features into $\mathbf{H}$ through gated Cross-Mamba, enabling accurate and reliable global feature fusion.

We detail the feature aggregation process between $\mathbf{M}_{\text{m}}$ and $\mathbf{H}$, which can be formally expressed as follows:

$$\mathbf{F} = \text{S}^2\text{CA}(\mathbf{H}, \mathbf{M}_{\text{m}}) = \text{GA}\left(\text{CS}^2\text{C}(\mathbf{H}, \mathbf{M}_{\text{m}}), \mathbf{H}\right). \tag{19}$$

The CS^{2}C module performs center-aware spatial–spectral correction on $\mathbf{M}_{\text{m}}$ using sliding-window spatial–spectral cross-attention to generate the corrected reference $\mathbf{M}_{\text{c}}$, which can be formulated as follows:

$$\mathbf{M}_{\text{c}} = \text{CS}^2\text{C}(\mathbf{H}, \mathbf{M}_{\text{m}}) = \mathcal{C}_{spa}^{\mathbf{c}}(\mathcal{S}_{pe}^{\mathbf{c}}(\mathbf{M}_{\text{m}}, \mathcal{S}_{pe}^{\mathbf{s}}(\mathbf{H})), \mathcal{S}_{pe}^{\mathbf{s}}(\mathbf{H}))) \tag{20}$$

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where $\mathcal{S}^{\mathbf{s}}pe$ denotes spectral self-attention applied to $\mathbf{H}$ for spectral enhancement, $\mathcal{S}^{\mathbf{c}}pe$ denotes spectral cross-attention between $\mathbf{M}_{\mathrm{m}}$ and $\mathbf{H}$ for spectral correction of $\mathbf{M}_{\mathrm{m}}$, and $\mathcal{C}^{\mathbf{c}}_{spa}$ denotes the center-aware spatial correction (CSC) module that establishes intrapatch spatial correspondence between $\mathbf{M}_{\mathrm{m}}$ and $\mathbf{H}$ using center-aware sliding-window spatial cross-attention. The operation of $\mathcal{C}^{\mathbf{c}}_{spa}$ can be further defined as follows:

$$\mathcal{C}^{\mathbf{c}}_{spa}(\mathbf{H}, \mathbf{M}_{\mathrm{m}}) = \mathcal{C}\left(\mathcal{W}^{\mathbf{c}}_{spa}(\mathbf{H}, \mathbf{M}_{\mathrm{m}})\right) \tag{21}$$

where $\mathcal{W}^{\mathbf{c}}_{spa}$ performs spatial cross-attention within each sliding window between $\mathbf{H}$ and $\mathbf{M}$m, and $\mathcal{C}$ applies a cosine window filtering to the attention outputs to achieve center-aware weighting, thereby alleviating boundary discontinuities caused by previous patch matching.

The GA module with learnable gated Cross-Mamba can dynamically control texture transfer from the corrected reference $\mathbf{M}_{\mathrm{c}}$ to the target $\mathbf{H}$. This adaptive mechanism selectively aggregates global features while suppressing erroneous artifacts

$$\mathbf{F} = \mathrm{GA}(\mathbf{H}, \mathbf{M}_{\mathrm{c}}) = \mathbf{H} + \mathcal{G}(\mathcal{F}(\mathbf{H}, \mathbf{M}_{\mathrm{c}}), \mathbf{H}) \tag{22}$$

where $\mathcal{F}(\cdot)$ denotes a Cross-Mamba, $\mathcal{G}(\cdot)$ implements the gating function with spectral–spatial awareness. The residual connection preserves original $\mathbf{H}$ features. The gating weights are learned through

$$\mathcal{G} = \sigma(\mathcal{L}(\mathcal{P}(cat[\mathbf{H}, \mathcal{F}(\mathbf{H}, \mathbf{M}_{\mathrm{c}})]))) \tag{23}$$

where *cat* performs feature concatenation along both channel and spatial dimensions to facilitate subsequent spectral–spatial aware gating, $\mathcal{P}$ combines max and average pooling, $\mathcal{L}$ is a linear layer, and σ denotes the sigmoid activation, enabling soft feature selection.

The S^{2}CA module performs center-aware spectral–spatial correction followed by global selective aggregation, preventing erroneous information transfer from the matched reference $\mathbf{M}_{\mathrm{m}}$ to the target $\mathbf{H}$ while enhancing spatial details and preserving spectral fidelity.

The aggregated features $\mathbf{F}$ from $\mathbf{H}$ and $\mathbf{M}_{\mathrm{m}}$ are injected layer-by-layer into the diffusion denoising backbone, which consists of multiple GA modules, to produce the SR result at each time step. This precise aggregation enables the reconstruction of high-quality HR-HSI for unpaired HSI-SR.

IV. Experiment

A. Dataset Description

To evaluate the proposed unpaired HSI-SR method, three simulated unpaired datasets were constructed from five public datasets. In particular, a cross-scene Pavia dataset was built using the Pavia Center dataset and Pavia University dataset, a cross-temporal Houston dataset was formed from the Houston13 dataset and Houston18 dataset, and a cross-region ChikuseiR dataset was generated by spatially partitioning the Chikusei dataset.

Following the Wald protocol, the original HR-HSIs are treated as GT. LR-HSI are generated by applying Gaussian blurring followed by 4× spatial downsampling, while HR-MSI are obtained via spectral degradation using the corresponding spectral response functions.

To simulate realistic unpaired scenarios, a cross-domain matching strategy is adopted. For each LR-HSI sample, an HR-MSI patch from a different scene, time, or region is retrieved based on the highest structural similarity index (SSIM). This ensures that the paired inputs remain spatio-temporally inconsistent while preserving certain structural similarity.

1) Pavia Dataset (Cross-Scene): The Pavia Center dataset is divided into 12 blocks of size 160 × 320 × 102, with nine used for training and three for testing. Each block is cropped into HR-HSI patches of size 160 × 160 × 102, which serve as GT. Corresponding LR-HSI patches are generated via Gaussian blurring and 4× downsampling.

To simulate cross-scene data, HR-MSI is derived from the Pavia University dataset via spectral degradation. For each LR-HSI patch, the HR-MSI patch with the highest SSIM is selected to form the unpaired input pair.

2) Houston Dataset (Cross-Temporal): The Houston18 dataset is used as the hyperspectral source. A region of size 480 × 1280 × 48 is divided into eight blocks of size 480 × 160 × 48, with six for training and two for testing. From each block, HR-HSI patches of size 160 × 160 × 48 are extracted as GT, and LR-HSI patches of size 40 × 40 × 48 are generated accordingly.

The HR-MSI reference patches are retrieved from the Houston13 dataset, simulating the fusion of data acquired at different times.

3) ChikuseiR Dataset (Cross-Region): The central region of the Chikusei dataset (2048 × 2048 × 128) is split into left and right halves. The right half is further divided into 16 blocks of size 512 × 256 × 128, with 12 used for training and four for testing. HR-HSI patches of size 256 × 256 × 128 are cropped as GT, and LR-HSI patches of size 64 × 64 × 128 are generated via Gaussian blurring and 4× downsampling.

To simulate cross-region conditions, the left half serves as the HR-MSI source. For each LR-HSI patch from the right region, the HR-MSI patch with the highest SSIM is selected to form the unpaired input pair.

B. Comparison Methods

Since research on unpaired HSI-SR remains limited, we construct a comprehensive evaluation framework to compare the proposed method with three categories of state-of-the-art (SOTA) approaches. First, considering that unpaired HSI-SR targets scenarios where LR-HSI lacks spatio-temporally consistent HR-MSI, we adopt single HSI-SR methods that do not require reference images as baseline comparisons, including SCPSN [36] and CST [37]. Second, to analyze the impact of spatio-temporally inconsistent image pairs in fusion-based methods, we include representative fusion-based HSI-SR approaches, namely PanMamba [38] and ARConv [39]. In addition, several representative Ref-SR methods from the natural image domain are introduced, including TTSR [13], MASA-SR [14], C^2 [40], and DATSR [41]. Their architectures are retained while adapting the input dimensions

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TABLE I

QUANTITATIVE COMPARISONS WITH SOTA APPROACHES ACROSS THREE DATASETS USING FIVE METRICS. THE SUFFIX (M) DENOTES THAT THE CORRESPONDING METHOD IS PREPROCESSED WITH OUR MATCHING MODULE. THE BEST RESULTS ARE INDICATED IN BOLD, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Dataset / Method	Pavia					Houston					ChikuseiR				
	PSNR	SAM	ERGAS	CC	RMSE	PSNR	SAM	ERGAS	CC	RMSE	PSNR	SAM	ERGAS	CC	RMSE
SCPSN	34.5741	4.6687	3.6922	0.9697	0.0189	38.3979	2.1935	2.1610	0.9845	0.0116	41.7732	1.6224	3.1406	0.9779	0.0072
CST	37.8263	3.5521	2.7184	0.9835	0.0130	44.3648	1.3981	1.1453	0.9948	**0.0043**	43.8451	1.6435	3.1782	0.9884	0.0053
PanMamba	34.1822	3.4983	3.7842	0.9765	0.0170	39.5516	2.1223	1.2940	0.9914	0.0075	39.4984	1.8902	3.7846	0.9667	0.0098
PanMamba(M)	35.0376	4.6101	3.4754	0.9735	0.0177	39.6317	2.0932	1.8872	0.9887	0.0101	41.0700	1.7615	3.3120	0.9752	0.0082
ARconv	35.4120	4.3747	3.4309	0.9748	0.0170	42.0650	1.5936	1.4459	0.9927	0.0077	41.5746	1.6737	3.3820	0.9757	0.0078
ARconv(M)	36.3930	4.2282	3.0895	0.9787	0.0152	42.5378	1.5101	1.3712	0.9935	0.0073	42.0918	1.6143	3.0921	0.9787	0.0073
TTSR	36.4943	4.4518	2.5623	0.9810	0.0147	43.0318	1.9937	1.3798	0.9909	0.0057	43.4569	1.6987	3.1498	0.9889	0.0055
MASA-SR	38.1695	3.6019	2.5418	0.9894	0.0110	44.9422	1.0903	0.9834	0.9936	0.0047	43.2898	1.5090	2.8534	0.9922	0.0055
C^2	37.4436	3.6630	2.7549	0.9869	0.0118	44.2622	0.9605	1.1476	0.9949	0.0050	37.9059	2.0144	4.0651	0.9584	0.0139
DATSR	37.8263	3.5521	2.7184	0.9835	0.0130	42.7057	1.3804	1.3327	0.9942	0.0071	43.6088	1.3897	2.5886	0.9842	0.0061
RGTGAN	39.9450	3.3149	2.1559	0.9895	0.0100	44.1359	1.1222	1.1592	0.9951	0.0062	45.8793	1.1602	2.1782	0.9891	0.0046
Proposed	**41.7775**	**2.5772**	**1.7213**	**0.9935**	**0.0081**	**47.1707**	**0.5915**	**0.8153**	**0.9977**	**0.0043**	**50.2846**	**0.8173**	**1.6282**	**0.9943**	**0.0027**
Ideal Value	$+\infty$	0	0	1	0	$+\infty$	0	0	1	0	$+\infty$	0	0	1	0

to hyperspectral data, and all models are trained under a unified experimental configuration. Furthermore, RGTGAN [35], a Ref-SR method designed for remote sensing imagery, is also included for comparison. In addition to the original fusion-based HSI-SR methods, we further conduct experiments using the proposed matching module to preprocess unpaired reference images before feeding them into the fusion network. The module enhances local similarity between the reference and target images, although no spatial correspondence exists between them. This setting is designed to investigate the effect of enhanced local similarity on fusion-based HSI-SR under unpaired conditions, rather than to validate the matching module itself.

C. Evaluation Criteria and Implementation Details

Five widely used metrics are employed to quantitatively evaluate reconstruction performance, including peak signal-to-noise ratio (PSNR), correlation coefficient (CC), spectral angle mapper (SAM), relative dimensionless global error in synthesis (ERGAS), and root-mean-square error (RMSE). Higher PSNR and CC indicate better reconstruction quality, whereas lower SAM, ERGAS, and RMSE reflect smaller spectral and spatial distortions.

All experiments are implemented in PyTorch and conducted on an NVIDIA GTX 3090 GPU. The model is trained using the Adam optimizer for 1000 epochs with a batch size of 2 and an initial learning rate of 1×10^{-4}. The diffusion process uses $T = 1000$ time steps. A linear noise schedule is adopted, where β_t increases from $\beta_1 = 1 \times 10^{-5}$ to $\beta_T = 0.002$. During testing, the DDIM sampling strategy is used to reduce the number of sampling steps to 50, balancing reconstruction quality and inference efficiency. In the C^2PM module, three matching operations with patch sizes of 4×4, 8×8, and 16×16 are performed to generate multiscale matched patches for reconstruction. The number of clustering categories is set to match the land-cover classes in each dataset, with 9, 15, and 19 categories for the Pavia, Houston, and Chikusei datasets.

D. Experimental Results

1) Quantitative Evaluations: Table I reports quantitative results on the Pavia dataset, Houston dataset, and ChikuseiR dataset. The proposed C^2MA-Diff-Net consistently outperforms all SOTA methods across all evaluation metrics, achieving the highest PSNR and CC as well as the lowest ERGAS, RMSE, and SAM. These results demonstrate its strong capability to simultaneously recover spatial details while preserving spectral fidelity in the unpaired HSI-SR task.

Compared with single HSI-SR methods, C^2MA-Diff-Net effectively exploits texture information from unpaired HR-MSI to enhance spatial reconstruction. Compared with fusion-based methods, it can search for and selectively transfer informative textures even without spatial correspondence. Furthermore, compared with Ref-SR methods, our approach performs scale-co-occurrence-assisted matching and adaptive dynamic aggregation to reliably discover and transfer texture details from reference images. The additional experiments with the matching-module-preprocessed fusion network achieve better superresolution performance than the original fusion network, indicating that under unpaired settings, improving the local structural consistency of the reference image has a positive impact on the reconstruction performance of fusion-based frameworks.

2) Qualitative Evaluations: We randomly select one image from the test set of each of the three datasets to visualize the

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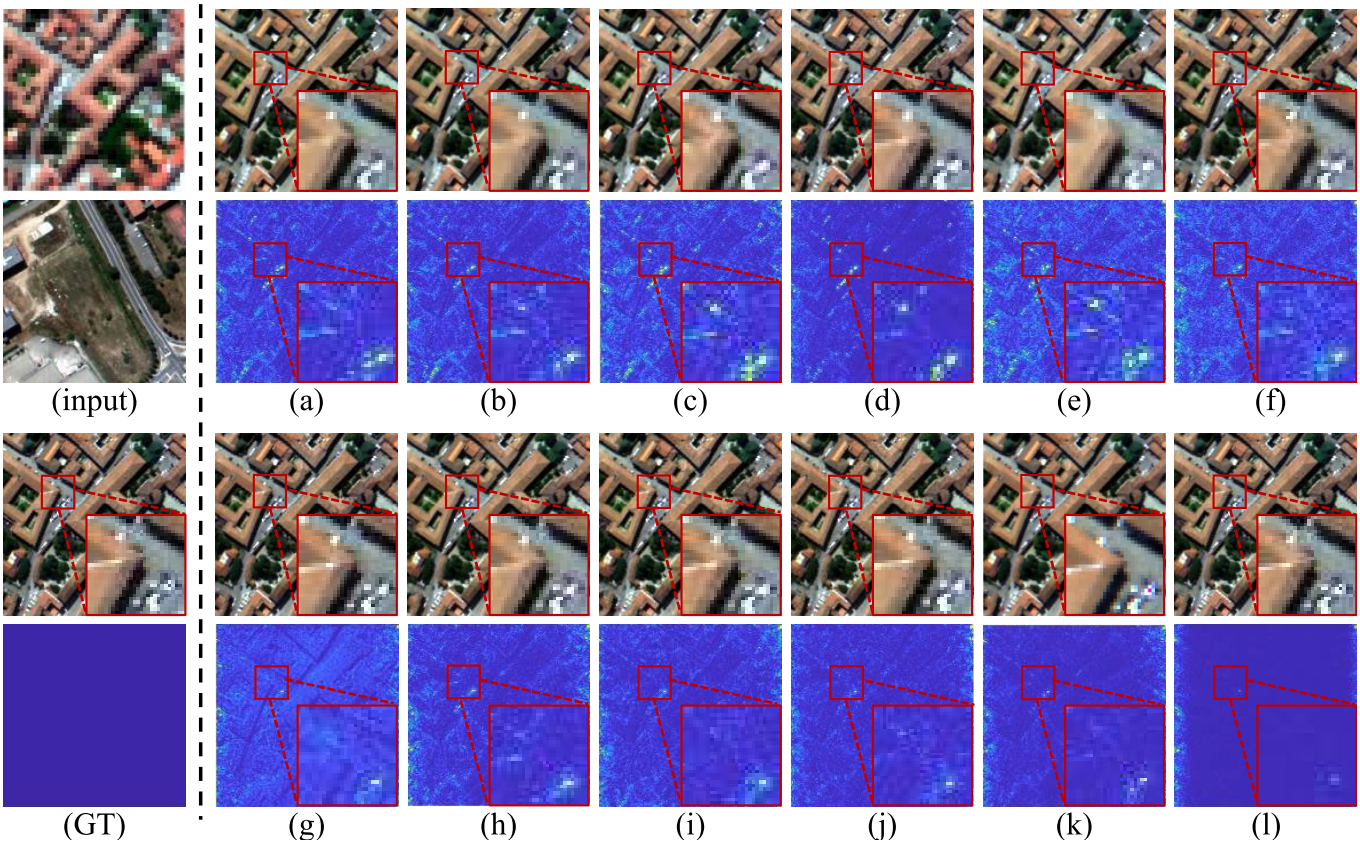


Fig. 4. Visual results and MAE maps of different HSI-SR methods on the Pavia dataset. Leftmost column: top shows input unpaired LR-HSI and HR-MSI pair; bottom shows GT and corresponding MAE map. Others. (a) SCPSN. (b) CST. (c) PanMamba. (d) PanMamba(M). (e) ARconv. (f) ARconv(M). (g) TTSR. (h) MASA-SR. (i) C^2. (j) DATSR. (k) RGTGAN. (l) Proposed. The suffix (M) denotes that the corresponding method is preprocessed with our matching module.

pseudocolor superresolution results and their corresponding residual maps. The residuals are calculated as the mean absolute error (MAE) between the reconstructed images and the reference images. In Figs. 4–6, the leftmost column shows the input unpaired HR-MSI and LR-HSI image pair at the top, and GT with the corresponding MAE map at the bottom. The remaining columns show the results of comparison methods: the top row displays the reconstructed images, and the bottom row shows the residual maps relative to GT. (a)–(k) correspond to the comparison methods, respectively, and (l) denotes the results of our proposed method.

As shown in Figs. 4–6, in contrast, our method achieves the visually sharpest and most faithful reconstruction results in three datasets. It not only enhances spatial details but also improves spectral fidelity. The corresponding residual maps are consistently the darkest, with the fewest discernible details, and are closest to the residual map of the GT, reflecting the smallest reconstruction errors. This qualitatively demonstrates that our method achieves the best performance.

The enlarged local regions further demonstrate the effectiveness of the proposed method in recovering fine-grained details under unpaired settings. As shown in Fig. 4, our method better preserves the structural continuity and boundary details of urban buildings. In Fig. 5, our method reconstructs clearer road edges and more continuous urban textures. In Fig. 6, our method generates more regular farmland textures with fewer distortions and artifacts. The MAE maps consistently exhibit lower reconstruction errors in these challenging regions.

E. Ablation Study

1) Effectiveness of C^2PM: To evaluate the effectiveness of the C^2PM module and its dual-branch design in unpaired HSI-SR, three ablation settings are considered: *1) w/o C^2PM:* removing the entire module and directly feeding HR-MSI and LR-HSI into the S^2CA module; *2) w/o RM:* removing the reference-based matching branch, where the network relies only on internal self-similarity within LR-HSI; and *3) w/o SM:* removing the scale-co-occurrence self-matching branch, where transferable structures are retrieved only from the RM branch.

As shown in Fig. 7, removing either branch or the entire module leads to noticeable performance degradation, with the most severe drop observed in w/o C^2PM.

This degradation occurs because unpaired images inherently exhibit spatial mismatches. Without reliable matching, reference images may introduce spatial offsets that disrupt spectral–spatial consistency. The performance drop in w/o RM highlights the importance of external high-resolution textures for detail reconstruction, while the degradation in w/o SM indicates that internal structural priors provide complementary information.

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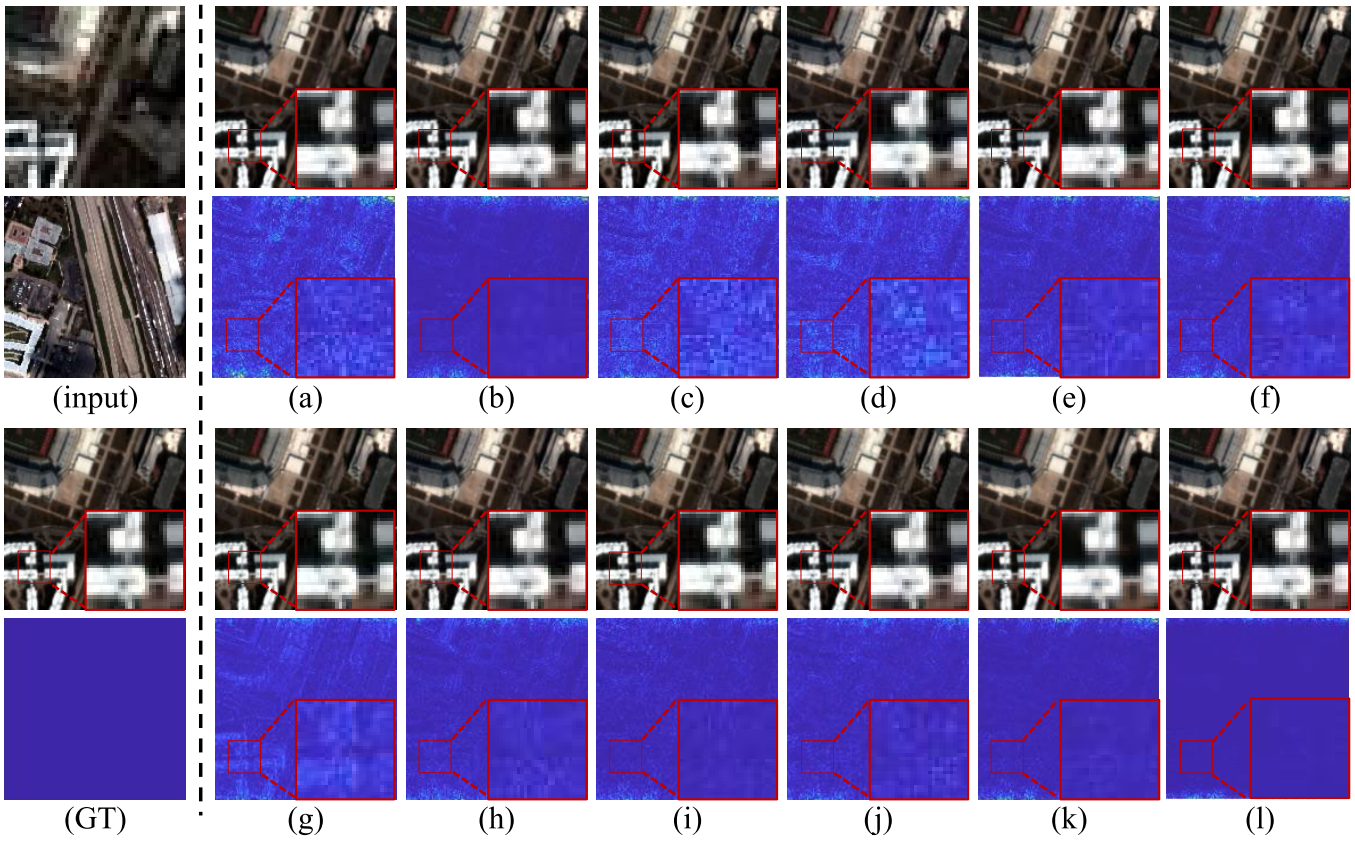


Fig. 5. Visual results and MAE maps of different HSI-SR methods on the Houston dataset. Leftmost column: top shows input unpaired LR-HSI and HR-MSI pair; bottom shows GT and corresponding MAE map. Others. (a) SCPSN. (b) CST. (c) PanMamba. (d) PanMamba(M). (e) ARconv. (f) ARconv(M). (g) TTSR. (h) MASA-SR. (i) C^2. (j) DATSR. (k) RGTGAN. (l) Proposed. The suffix (M) denotes that the corresponding method is preprocessed with our matching module.

These results demonstrate that C^2PM effectively establishes reliable correspondences between unpaired images. By combining external texture transfer with self-scale-co-occurrence assistance, the dual-branch design enables more reliable reconstruction.

2) Effectiveness of S^2Ca: To analyze the role of S^2CA, two ablation settings are considered: *1) w/o CS^2C:* removing the CS^2C module; and *2) w/o GA:* replacing the GA module with a single Mamba module.

As shown in Fig. 8, removing either component leads to noticeable performance degradation, while the full model consistently achieves the best results. Removing CS^2C severely degrades both spatial and spectral metrics, highlighting its role in spatial–spectral correction for reference images. Replacing GA with a standard Mamba structure also results in significant performance degradation, indicating that simple sequence modeling cannot effectively handle unreliable reference information. The gating mechanism enables reliability-aware weighting and selective aggregation, filtering out irrelevant or erroneous textures.

3) Effectiveness of Diffusion: To verify the effectiveness of the diffusion model, two ablation experiments are conducted: 1) *w/o Diff:* removing the diffusion model; and 2) *conventional Diff:* replacing the proposed formulation with a conventional conditional diffusion model that concatenates the conditional image with the noisy input.

TABLE II
UNPAIRED HSI-SR RESULTS AMONG NONDIFFUSION, CONVENTIONAL CONDITIONAL DIFFUSION AND OUR PROPOSED METHOD

Dataset	Methods	PSNR ↑	SAM ↓	ERGAS ↓	RMSE ↓
Pavia	w/o Diff	40.0571	3.4314	2.1536	0.0091
	conventional Diff	39.0770	3.6388	2.3938	0.0110
	Proposed	**41.7775**	**2.5772**	**1.7213**	**0.0081**
Houston	w/o Diff	46.0118	0.9054	0.9969	0.0049
	conventional Diff	43.7137	1.2989	0.9951	0.0064
	Proposed	**47.1707**	**0.5915**	**0.8153**	**0.0043**
ChikuseiR	w/o Diff	48.2225	1.0737	0.9916	0.0035
	conventional Diff	46.7739	1.1891	0.9902	0.0042
	Proposed	**50.2846**	**0.8173**	**1.6282**	**0.0027**

As shown in Table II, removing the diffusion model leads to severe degradation in reconstruction quality, demonstrating its importance for modeling high-frequency details and maintaining structural stability. The diffusion process progressively reconstructs fine textures through iterative denoising, which alleviates instability caused by unreliable references in unpaired scenarios. Using a conventional conditional diffusion model also results in inferior performance, indicating that simple condition concatenation cannot effectively bridge

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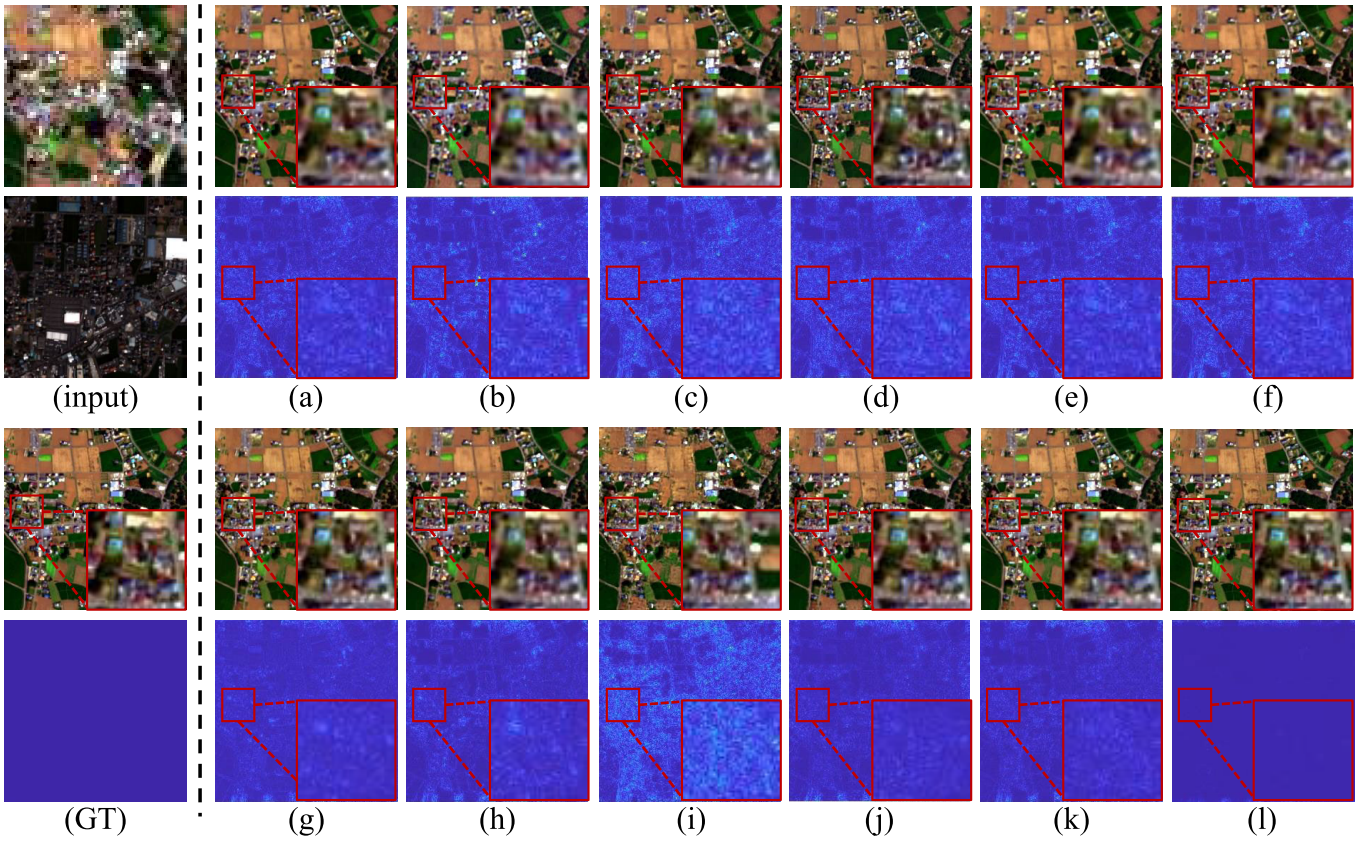


Fig. 6. Visual results and MAE maps of different HSI-SR methods on the ChikuseiR dataset. Leftmost column: top shows input unpaired LR-HSI and HR-MSI pair; bottom shows GT and corresponding MAE map. Others. (a) SCPSN. (b) CST. (c) PanMamba. (d) PanMamba(M). (e) ARconv. (f) ARconv(M). (g) TTSR. (h) MASA-SR. (i) C^2. (j) DATSR. (k) RGTGAN. (l) Proposed. The suffix (M) denotes that the corresponding method is preprocessed with our matching module.

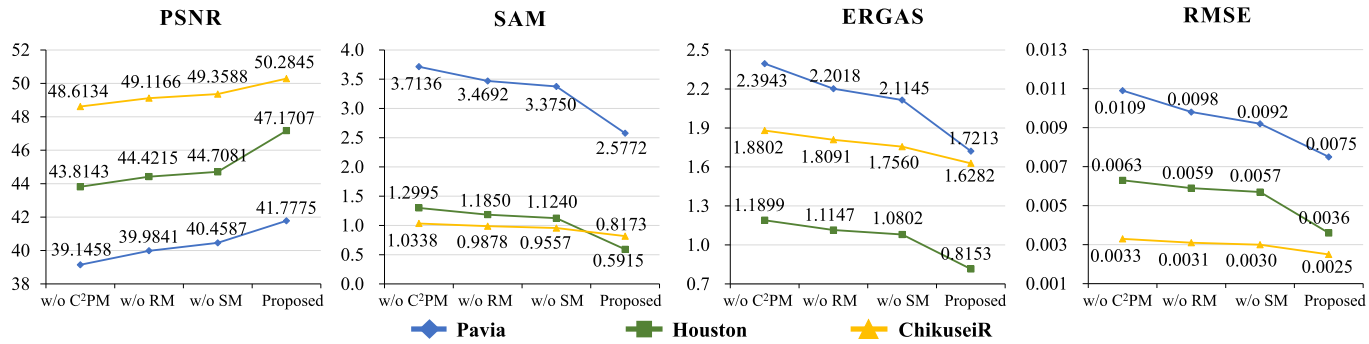


Fig. 7. Ablation study results of the C^2PM module of the proposed method, including its reference-based matching branch and self-scale-co-occurrence matching branch, on the Pavia, Houston, and ChikuseiR datasets.

cross-image distribution gaps in unpaired HSI-SR. On the contrary, the proposed formulation integrates conditional guidance with progressive reconstruction, enabling more reliable detail recovery and improved reconstruction stability. Overall, these results verify the effectiveness of the proposed diffusion framework for unpaired HSI-SR.

F. Computational Complexity Analysis

As shown in Table III, representative single HSI-SR, fusion-based HSI-SR, and Ref-SR methods are selected for computational complexity comparison. All methods are evaluated under the same settings on the same hardware platform using their respective optimal configurations. The comparison includes model parameters, FLOPs, and inference runtime.

The proposed C^2MA-Diff-Net contains only 3.6351 M parameters, which is the lowest among all compared methods while maintaining the best reconstruction performance. In terms of computational cost, the proposed method requires 37.14 G FLOPs, which is lower than that of RGTGAN and comparable to several methods. The proposed method achieves an average inference runtime of 31.967 ms per diffusion

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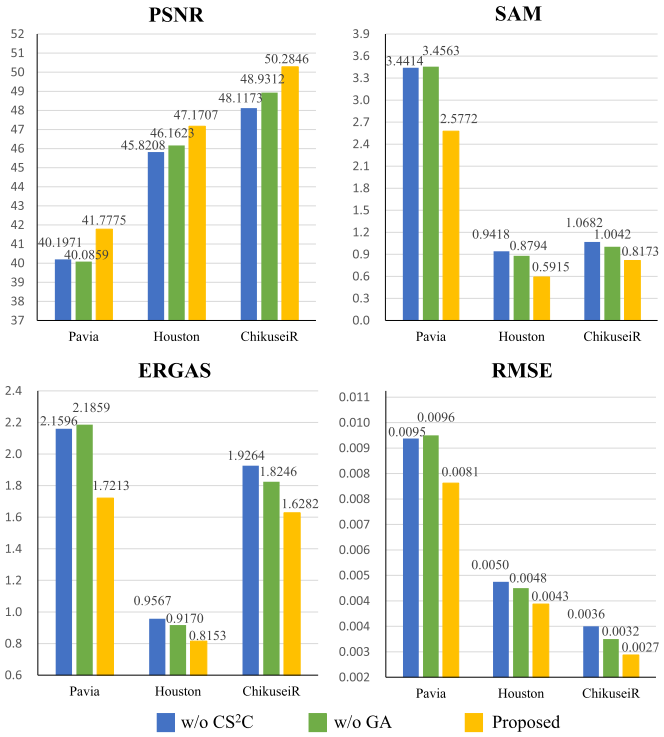


Fig. 8. Ablation study results of the S²CA module, including CS²C and GA, on the Pavia, Houston, and ChikuseiR datasets.

TABLE III
COMPARISON OF COMPUTATIONAL COMPLEXITY

Methods	Parameters(M)	Flops(G)	Times(ms)
CST	11.1967	22.16	47.729
ARconv	15.9476	25.25	74.570
Panmamba	3.7074	43.64	56.137
MASA-SR	5.0426	35.88	17.079
RGTGAN	23.6449	215.53	82.113
Proposed	3.6351	37.14	31.967

sampling step. Although the iterative denoising process introduces additional inference overhead due to multiple sampling steps, the DDIM sampling strategy effectively reduces the sampling cost while preserving satisfactory reconstruction quality. Overall, these results demonstrate that the proposed C²MA-Diff-Net achieves a favorable balance between reconstruction performance and computational complexity.

G. Impact of Reference Quality

To investigate the impact of the similarity between the reference and target image on the proposed method, we conducted additional experiments under different reference selection strategies. In particular, two settings were considered: 1) SSIM-based retrieval, where reference images with the highest SSIM to the target LR-HSI are selected; and 2) random selection, where reference images are randomly selected. All reference images are drawn from datasets captured at different acquisition times or scenes relative to the target LR-HSI.

TABLE IV
PERFORMANCE COMPARISON UNDER DIFFERENT REFERENCE SELECTION STRATEGIES. "RETRIEVED" DENOTES SSIM-BASED RETRIEVAL

Dataset	Reference Type	PSNR ↑	SAM ↓	ERGAS ↓	RMSE ↓
Pavia	Retrieved	41.7775	2.5772	1.7213	0.0081
	Random	41.1676	3.0185	1.9103	0.0085
Houston	Retrieved	47.1707	0.5915	0.8153	0.0043
	Random	46.8811	0.7416	0.8507	0.0044
ChikuseiR	Retrieved	50.2846	0.8173	1.6282	0.0027
	Random	49.8631	0.8797	1.6697	0.0028

As shown in Table IV, the performance of the proposed method degrades under the random selection setting. This indicates that reference quality affects unpaired HSI-SR performance and suggests that, under practical spatio-temporal inconsistencies, selecting reference images with structural and textural characteristics as similar as possible to those of the target LR-HSI can provide more effective texture information and lead to better HSI-SR performance. Moreover, even with randomly selected references, our method still outperforms the comparison methods using retrieved references.

V. CONCLUSION

In this article, we propose a C²MA-Diff-Net for unpaired HSI-SR, which takes cross-scene, cross-temporal, or cross-region unpaired LR-HSI and HR-MSI as inputs. Breaking the reliance on spatio-temporal consistency between input images, the proposed method significantly expands the applicability and practical value of HSI-SR. C²MA-Diff-Net is built upon a conditional diffusion framework, which progressively reconstructs HR-HSI through iterative optimization with scale-co-occurrence-assisted category-guided patch matching and adaptive aggregation. Our proposed C²PM module performs CGPM with the assistance of self-scale-co-occurrence, efficiently identifying texture-similar patches from HR-MSI while mining cross-scale complementary information from LR-HSI itself. Our proposed S²CA module performs reliable feature aggregation while maintaining spectral fidelity by rectifying reference features with center-aware sliding-window spectral–spatial attention and selectively integrating textures via adaptive gated Cross-Mamba. We compare C²MA-Diff-Net with SOTA approaches on three datasets, where quantitative and qualitative results demonstrate its superior performance for unpaired HSI-SR, validating the effectiveness of our method.

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